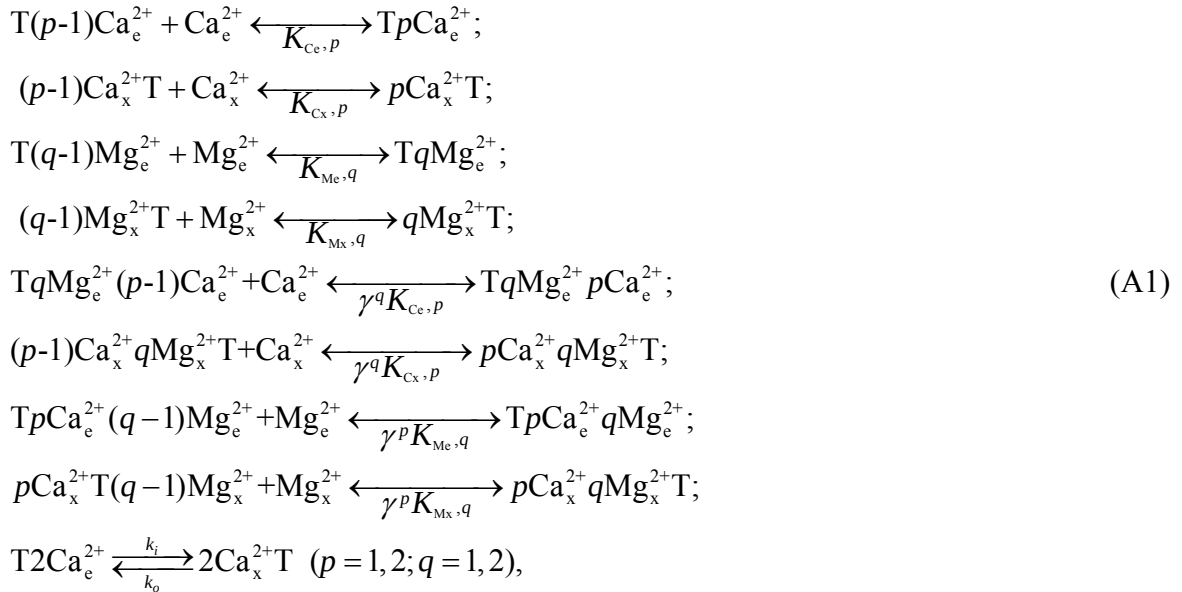


# Supporting Materials for “Characterization of $\text{Mg}^{2+}$ Inhibition of Mitochondrial $\text{Ca}^{2+}$ Uptake by a Mechanistic Model of Mitochondrial $\text{Ca}^{2+}$ Uniporter”.

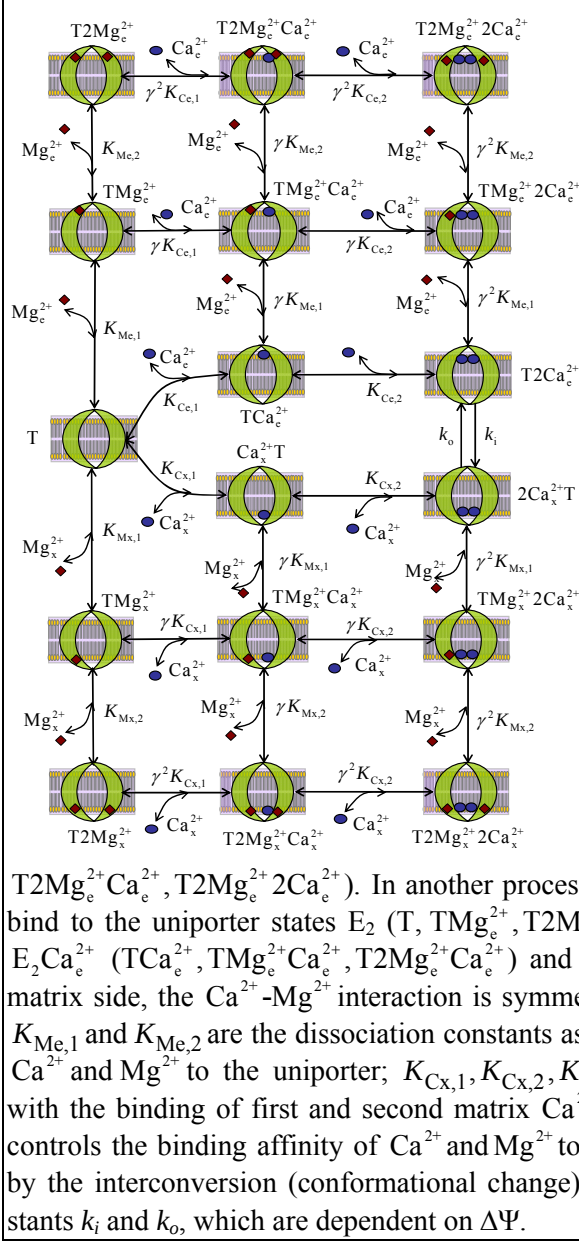
Ranjan K. Pradhan, Feng Qi, Daniel A. Beard, and Ranjan K. Dash\*  
Biotechnology and Bioengineering Center and Department of Physiology, Medical College of Wisconsin, Milwaukee, WI-53226. \*Corresponding author ([rdash@mcw.edu](mailto:rdash@mcw.edu))

## Appendix SA: Derivations of Mitochondrial $\text{Ca}^{2+}$ Uniporter Flux Expressions with the Proposed $\text{Mg}^{2+}$ Inhibition Mechanism

This appendix presents the detailed derivations of the kinetic models of mitochondrial  $\text{Ca}^{2+}$  uniporter, based on the proposed kinetic mechanism of  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  transport via the uniporter (see Figure S1). Assuming a quasi-steady state flux and rapid equilibrium binding of external and internal  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter, the binding and translocation reactions for the uniporter with presumed  $2\text{Ca}^{2+}$  and  $2\text{Mg}^{2+}$  binding sites can be written as



where  $K_{\text{Ce},p}$ ,  $K_{\text{Cx},p}$ ,  $K_{\text{Me},q}$  and  $K_{\text{Mx},q}$  are the true dissociation constants associated with the binding of external and internal  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter;  $\gamma$  is a control parameter ( $1 \leq \gamma \leq \infty$ ) that determines the affinities of  $\text{Ca}^{2+}$  ( $\text{Mg}^{2+}$ ) binding to the  $\text{Mg}^{2+}$  ( $\text{Ca}^{2+}$ ) bound uniporter states;  $k_i$  and  $k_o$  are the forward and reverse rate constants involved in the interconversion of uniporter complexes  $\text{T}2\text{Ca}_e^{2+}$  and  $2\text{Ca}_x^{2+}\text{T}$ . The interconversion process of these two complex states involves translocation of positive charges across the inner mitochondrial membrane (IMM), and hence these rate constants are functions of membrane potential ( $\Delta\Psi$ ). Furthermore, depending upon the physical locations of the  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  binding sites on the uniporter, the dissociation constants  $K_{\text{Ce},p}$ ,  $K_{\text{Cx},p}$ ,  $K_{\text{Me},q}$  and  $K_{\text{Mx},q}$  can also depend on  $\Delta\Psi$ .



**Figure S1: Proposed kinetic mechanism of  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  transport via the  $\text{Ca}^{2+}$  uniporter.** A linear mixed-type inhibition mechanism is proposed for the  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  transport. Shown are the schematics of different cytosolic and matrix  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  bound uniporter states during the binding and translocation of  $\text{Ca}^{2+}$  via the uniporter. The uniporter functional unit is assumed to have two binding sites for  $\text{Ca}^{2+}$  and two distinct allosteric sites for  $\text{Mg}^{2+}$ , facing either the cytosolic side or the matrix side depending on the state of the uniporter. In one process, two  $\text{Ca}^{2+}$  ions from the cytosolic side cooperatively bind to the uniporter T in two different steps to form the complexes  $\text{TCa}_e^{2+}$  and  $\text{T2Ca}_e^{2+}$ . The complex  $\text{T2Ca}_e^{2+}$  then undergoes conformational changes to form the complex  $2\text{Ca}_x^{2+}\text{T}$  in the matrix side. In a reverse process, the complex  $2\text{Ca}_x^{2+}\text{T}$  dissociates in two different steps to form the complex  $\text{Ca}_x^{2+}\text{T}$ , unbound uniporter T, and two free  $\text{Ca}^{2+}$  ions in the matrix side. For  $\text{Ca}^{2+}$ - $\text{Mg}^{2+}$  interactions, two  $\text{Mg}^{2+}$  ions from the cytosolic side cooperatively bind to the uniporter states  $\text{E}_1$  (T,  $\text{TCa}_e^{2+}$ ,  $\text{T2Ca}_e^{2+}$ ) in two different steps to form the complexes  $\text{E}_1\text{Mg}_e^{2+}$  ( $\text{TMg}_e^{2+}$ ,  $\text{TMg}_e^{2+}\text{Ca}_e^{2+}$ ,  $\text{TMg}_e^{2+}2\text{Ca}_e^{2+}$ ) and  $\text{E}_12\text{Mg}_e^{2+}$  ( $\text{T2Mg}_e^{2+}$ ,  $\text{T2Mg}_e^{2+}\text{Ca}_e^{2+}$ ,  $\text{T2Mg}_e^{2+}2\text{Ca}_e^{2+}$ ).

In another process, two  $\text{Ca}^{2+}$  ions from the cytosolic side can cooperatively bind to the uniporter states  $\text{E}_2$  (T,  $\text{TMg}_e^{2+}$ ,  $\text{T2Mg}_e^{2+}$ ) in two different steps to form the same complexes  $\text{E}_2\text{Ca}_e^{2+}$  ( $\text{TCa}_e^{2+}$ ,  $\text{TMg}_e^{2+}\text{Ca}_e^{2+}$ ,  $\text{T2Mg}_e^{2+}\text{Ca}_e^{2+}$ ) and  $\text{E}_22\text{Ca}_e^{2+}$  ( $\text{T2Ca}_e^{2+}$ ,  $\text{TMg}_e^{2+}2\text{Ca}_e^{2+}$ ,  $\text{T2Mg}_e^{2+}2\text{Ca}_e^{2+}$ ). In the matrix side, the  $\text{Ca}^{2+}$ - $\text{Mg}^{2+}$  interaction is symmetrical to that in the cytoplasmic side. Here  $K_{\text{Ce},1}$ ,  $K_{\text{Ce},2}$ ,  $K_{\text{Me},1}$  and  $K_{\text{Me},2}$  are the dissociation constants associated with the binding of first and second cytoplasmic  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter;  $K_{\text{Cx},1}$ ,  $K_{\text{Cx},2}$ ,  $K_{\text{Mx},1}$  and  $K_{\text{Mx},2}$  are the dissociation constants associated with the binding of first and second matrix  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter;  $1 \leq \gamma \leq \infty$  is a parameter that controls the binding affinity of  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter. The translocation of  $\text{Ca}^{2+}$  across the IMM by the interconversion (conformational change) reaction  $\text{T2Ca}_e^{2+} \leftrightarrow 2\text{Ca}_x^{2+}\text{T}$  is limited by two rate constants  $k_i$  and  $k_o$ , which are dependent on  $\Delta\Psi$ .

Under quasi-steady state, rapid equilibrium binding assumptions, the various states of the uniporter can be expressed as follows:

$$\begin{aligned}
 [\text{TpCa}_e^{2+}] &= \frac{[\text{Ca}^{2+}]_e^p}{\prod_{i=1}^p K_{\text{Ce},i}} [\text{T}]; \quad [p\text{Ca}_x^{2+}\text{T}] = \frac{[\text{Ca}^{2+}]_x^p}{\prod_{i=1}^p K_{\text{Cx},i}} [\text{T}]; \quad [\text{TpCa}_e^{2+}q\text{Mg}_e^{2+}] = \frac{[\text{Ca}^{2+}]_e^p [\text{Mg}^{2+}]_e^q}{\gamma^{p \times q} \cdot \prod_{i=1}^p K_{\text{Ce},i} \cdot \prod_{j=1}^q K_{\text{Me},j}} [\text{T}]; \\
 [\text{TqMg}_e^{2+}] &= \frac{[\text{Mg}^{2+}]_e^q}{\prod_{j=1}^q K_{\text{Me},j}} [\text{T}]; \quad [q\text{Mg}_x^{2+}\text{T}] = \frac{[\text{Mg}^{2+}]_x^q}{\prod_{j=1}^q K_{\text{Mx},j}} [\text{T}]; \quad [p\text{Ca}_x^{2+}q\text{Mg}_x^{2+}\text{T}] = \frac{[\text{Ca}^{2+}]_x^p [\text{Mg}^{2+}]_x^q}{\gamma^{p \times q} \cdot \prod_{i=1}^p K_{\text{Cx},i} \cdot \prod_{j=1}^q K_{\text{Mx},j}} [\text{T}],
 \end{aligned} \tag{A2}$$

where the concentrations of different uniporter states are expressed with respect to matrix volume;  $[Ca^{2+}]_e$ ,  $[Mg^{2+}]_e$ ,  $[Ca^{2+}]_x$  and  $[Mg^{2+}]_x$  represent the extra-matrix and matrix concentrations of  $Ca^{2+}$  and  $Mg^{2+}$ . The dissociation constants  $K_{Ce,p}$ ,  $K_{Cx,p}$ ,  $K_{Me,q}$  and  $K_{Mx,q}$  are in the units of molar concentration. Since the total concentration of the uniporter is constant, we have

$$[T]_{tot} = [T] + \sum_{p=1}^2 [TpCa_e^{2+}] + \sum_{p=1}^2 [pCa_x^{2+}T] + \sum_{q=1}^2 [TqMg_e^{2+}] + \sum_{q=1}^2 [qMg_x^{2+}T] \\ + \sum_{p=1}^2 \sum_{q=1}^2 [TpCa_e^{2+}qMg_e^{2+}] + \sum_{p=1}^2 \sum_{q=1}^2 [pCa_x^{2+}qMg_x^{2+}T]. \quad (A3)$$

By substituting Eq. (A3) into Eq. (A2), the concentration of unbound uniporter ( $[T]$ ) can be expressed in terms of total uniporter concentration ( $[T]_{tot}$ ) as

$$[T] = [T]_{tot} / D, \quad (A4)$$

where

$$D = 1 + \sum_{p=1}^2 \frac{[Ca^{2+}]_e^p}{\prod_{i=1}^p K_{Ce,i}} + \sum_{p=1}^2 \frac{[Ca^{2+}]_x^p}{\prod_{i=1}^p K_{Cx,i}} + \sum_{q=1}^2 \frac{[Mg^{2+}]_e^q}{\prod_{j=1}^q K_{Me,j}} + \sum_{q=1}^2 \frac{[Mg^{2+}]_x^q}{\prod_{j=1}^q K_{Mx,j}} \\ + \sum_{p=1}^2 \sum_{q=1}^2 \frac{[Ca^{2+}]_e^p [Mg^{2+}]_e^q}{\gamma^{p \times q} \cdot \prod_{i=1}^p K_{Ce,i} \cdot \prod_{j=1}^q K_{Me,j}} + \sum_{p=1}^2 \sum_{q=1}^2 \frac{[Ca^{2+}]_x^p [Mg^{2+}]_x^q}{\gamma^{p \times q} \cdot \prod_{i=1}^p K_{Cx,i} \cdot \prod_{j=1}^q K_{Mx,j}}. \quad (A5)$$

Based on the proposed kinetic mechanism (Figure S1), the net  $Ca^{2+}$  transport flux via the uniporter ( $J_{CU}$ ) can be written as

$$J_{CU} = k_i [T2Ca_e^{2+}] - k_o [2Ca_x^{2+}T] = \frac{[T]_{tot}}{D} \left( k_i \frac{[Ca^{2+}]_e^2}{\prod_{i=1}^2 K_{Ce,i}} - k_o \frac{[Ca^{2+}]_x^2}{\prod_{i=1}^2 K_{Cx,i}} \right). \quad (A6)$$

The flux expression for an uniporter with 2  $Ca^{2+}$  and 2  $Mg^{2+}$  binding sites consists of 2 pairs of dissociation constants for  $Ca^{2+}$  binding and 2 pairs of dissociation constants for  $Mg^{2+}$  binding ( $K_{Ce,p}$ ,  $K_{Cx,p}$ ,  $K_{Me,q}$ ,  $K_{Mx,q}$ ;  $p = 1,2$ ;  $q = 1,2$ ), two rate constants ( $k_i$ ,  $k_o$ ), and the control parameter  $\gamma$ , resulting in a total of eleven unknown parameters. The number unknown parameters are further reduced with the following assumptions regarding the binding of  $Ca^{2+}$  and  $Mg^{2+}$  to the uniporter.

**Model 1:** In the first variant of the model, the dissociation constants associated with the first binding steps of external and internal  $Ca^{2+}$  to the uniporter and the first binding steps of external and internal  $Mg^{2+}$  to the uniporter are assumed to be very large with the constraints:  $K_{Ce,1} K_{Ce,2} =$

$K_{Ce}^2$ ,  $K_{Cx,1}$   $K_{Cx,2} = K_{Cx}^2$ ,  $K_{Me,1}$   $K_{Me,2} = K_{Me}^2$ , and  $K_{Mx,1}$   $K_{Mx,2} = K_{Mx}^2$  are finite (full cooperativity). This approximation is valid when  $K_{Ce,1} \gg 1 \mu\text{M}$ ,  $K_{Ce,2} \ll 1 \mu\text{M}$ ,  $K_{Cx,1} \gg 1 \mu\text{M}$ ,  $K_{Cx,2} \ll 1 \mu\text{M}$ ,  $K_{Me,1} \gg 1 \text{mM}$ ,  $K_{Me,2} \ll 1 \text{mM}$ ,  $K_{Mx,1} \gg 1 \text{mM}$ , and  $K_{Mx,2} \ll 1 \text{mM}$ .

**Model 2:** In the second variant of the model, the dissociation constants associated with the binding of external and internal  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter are assumed to be equal to each other such that:  $K_{Ce,1} = K_{Ce,2} = K_{Ce}$ ,  $K_{Cx,1} = K_{Cx,2} = K_{Cx}$ ,  $K_{Me,1} = K_{Me,2} = K_{Me}$ , and  $K_{Mx,1} = K_{Mx,2} = K_{Mx}$  (partial cooperativity).

**Model 3:** In the third variant of the model, the dissociation constants associated with the binding of external and internal  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter are assumed to satisfy the constraints:  $K_{Ce,1} = K_{Ce}/2$ ,  $K_{Ce,2} = 2K_{Ce}$ ,  $K_{Cx,1} = K_{Cx}/2$ ,  $K_{Cx,2} = 2K_{Cx}$ ,  $K_{Me,1} = K_{Me}/2$ , and  $K_{Mx,2} = 2K_{Mx}$  (no cooperativity).

With these three different binding assumptions, the flux expression (A6) is reduced to

$$J_{CU} = \frac{[T]_{\text{tot}}}{D} \left( k_i \frac{[\text{Ca}^{2+}]_e^2}{K_{Ce}^2} - k_o \frac{[\text{Ca}^{2+}]_x^2}{K_{Cx}^2} \right), \quad (\text{A7})$$

where

$$D=D_1=1+\frac{[\text{Ca}^{2+}]_e^2}{K_{Ce}^2}+\frac{[\text{Ca}^{2+}]_x^2}{K_{Cx}^2}+\frac{[\text{Mg}^{2+}]_e^2}{K_{Me}^2}+\frac{[\text{Mg}^{2+}]_x^2}{K_{Mx}^2}+\frac{[\text{Ca}^{2+}]_e^2[\text{Mg}^{2+}]_e^2}{\gamma^4 K_{Ce}^2 \cdot K_{Me}^2}+\frac{[\text{Ca}^{2+}]_x^2[\text{Mg}^{2+}]_x^2}{\gamma^4 K_{Cx}^2 \cdot K_{Mx}^2}, \quad (\text{A8a})$$

$$D=D_2=1+\sum_{p=1}^2 \frac{[\text{Ca}^{2+}]_e^p}{K_{Ce}^p}+\sum_{p=1}^2 \frac{[\text{Ca}^{2+}]_x^p}{K_{Cx}^p}+\sum_{q=1}^2 \frac{[\text{Mg}^{2+}]_e^q}{K_{Me}^q}+\sum_{p=1}^2 \frac{[\text{Mg}^{2+}]_x^q}{K_{Mx}^q}+\sum_{p=1}^2 \sum_{q=1}^2 \frac{[\text{Ca}^{2+}]_e^p [\text{Mg}^{2+}]_e^q}{\gamma^{p \times q} \cdot K_{Ce}^p \cdot K_{Me}^q} \\ + \sum_{p=1}^2 \sum_{q=1}^2 \frac{[\text{Ca}^{2+}]_x^p [\text{Mg}^{2+}]_x^q}{\gamma^{p \times q} \cdot K_{Cx}^p \cdot K_{Mx}^q}, \quad (\text{A8b})$$

$$D=D_3=1+\sum_{p=1}^2 \frac{[\text{Ca}^{2+}]_e^p}{K_{Ce}^p / \prod_{i=1}^p \left( \frac{2-i+1}{i} \right)}+\sum_{p=1}^2 \frac{[\text{Ca}^{2+}]_x^p}{K_{Cx}^p / \prod_{i=1}^p \left( \frac{2-i+1}{i} \right)}+\sum_{q=1}^2 \frac{[\text{Mg}^{2+}]_e^q}{K_{Me}^q / \prod_{j=1}^q \left( \frac{2-j+1}{j} \right)}+\sum_{q=1}^2 \frac{[\text{Mg}^{2+}]_x^q}{K_{Mx}^q / \prod_{j=1}^q \left( \frac{2-j+1}{j} \right)} \\ + \sum_{p=1}^2 \sum_{q=1}^2 \frac{[\text{Ca}^{2+}]_e^p [\text{Mg}^{2+}]_e^q}{\gamma^{p \times q} \cdot K_{Ce}^p / \prod_{i=1}^p \left( \frac{2-i+1}{i} \right) \cdot K_{Me}^q / \prod_{j=1}^q \left( \frac{2-j+1}{j} \right)}+\sum_{p=1}^2 \sum_{q=1}^2 \frac{[\text{Ca}^{2+}]_x^p [\text{Mg}^{2+}]_x^q}{\gamma^{p \times q} \cdot K_{Cx}^p / \prod_{i=1}^p \left( \frac{2-i+1}{i} \right) \cdot K_{Mx}^q / \prod_{j=1}^q \left( \frac{2-j+1}{j} \right)}. \quad (\text{A8c})$$

Now the flux expression in each of these three versions of the model contains only four binding constants ( $K_{Ce}$ ,  $K_{Cx}$ ,  $K_{Me}$ ,  $K_{Mx}$ ), two rate constants ( $k_i$ ,  $k_o$ ), and one control parameter  $\gamma$ , for a total of seven unknown parameters. The denominators  $D_1$ ,  $D_2$  and  $D_3$  contains equal number of kinetic parameters;  $D_2$  and  $D_3$  look more complex than  $D_1$  due to the nature of the binding assumptions. Even if all three variant models are able to fit to the same experimental data, the estimated parameter values are likely to differ. In the absence of  $\text{Mg}^{2+}$  ( $[\text{Mg}^{2+}] = 0$ ), these flux expressions resemble to the flux expressions in our previous models of the uniporter (1, 2). Note that, Model

3 (no cooperativity) was not considered in our previous studies and included here as another possible case for the binding of  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  to the uniporter.

Under equilibrium conditions, the net  $\text{Ca}^{2+}$  transport flux via the uniporter is zero ( $J_{\text{CU}} = 0$ ). Therefore, the kinetic parameters  $K_{\text{Ce}}$ ,  $K_{\text{Cx}}$ ,  $k_i$  and  $k_o$  are further constrained by the following equilibrium relationship:

$$K_{\text{eq}} = \frac{k_i}{k_o} \cdot \frac{K_{\text{Cx}}^2}{K_{\text{Ce}}^2} = \left( \frac{[\text{Ca}^{2+}]_{\text{x}}^2}{[\text{Ca}^{2+}]_{\text{e}}^2} \right)_{\text{eq}}, \quad (\text{A9})$$

where  $K_{\text{eq}}$  is the equilibrium constant for the transport of  $2\text{Ca}^{2+}$  via the uniporter which is a function of  $\Delta\Psi$ . We also note here that the above kinetic parameters depend on  $\Delta\Psi$ .

## Appendix SB: Justification for the Proposed Mechanism of Pi Regulation of the Kinetic Parameters in the Model of Mitochondrial $\text{Ca}^{2+}$ Uniporter

This appendix provides a detailed model analysis justifying the proposed mechanism of Pi regulation of the kinetic parameters  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  (Eq. 9) in the present uniporter model. The kinetic mechanism by which Pi regulates the uniporter function (or kinetic parameters) is not clear from the available experimental data. On the other hand, without considering the proposed Pi regulation mechanism for  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$ , the model does not fit the available data with unique  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  values for all data sets, irrespective of the amounts of Pi present in the experimental buffer mediums, even with the proposed general linear mixed-type inhibition mechanism for  $\text{Mg}^{2+}$  inhibition of the uniporter function, indicating the role of Pi in regulating the uniporter function (or kinetic parameters). Therefore, we explored two possible fitting scenarios for parameterization of the model to best describe the available data. In one fitting scenario (Case A), only  $K_{\text{C}}^0$  was allowed to vary between different data sets, while  $K_{\text{M}}^0$  was fixed. In another fitting scenario (Case B), both  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  were allowed to vary between different data sets.

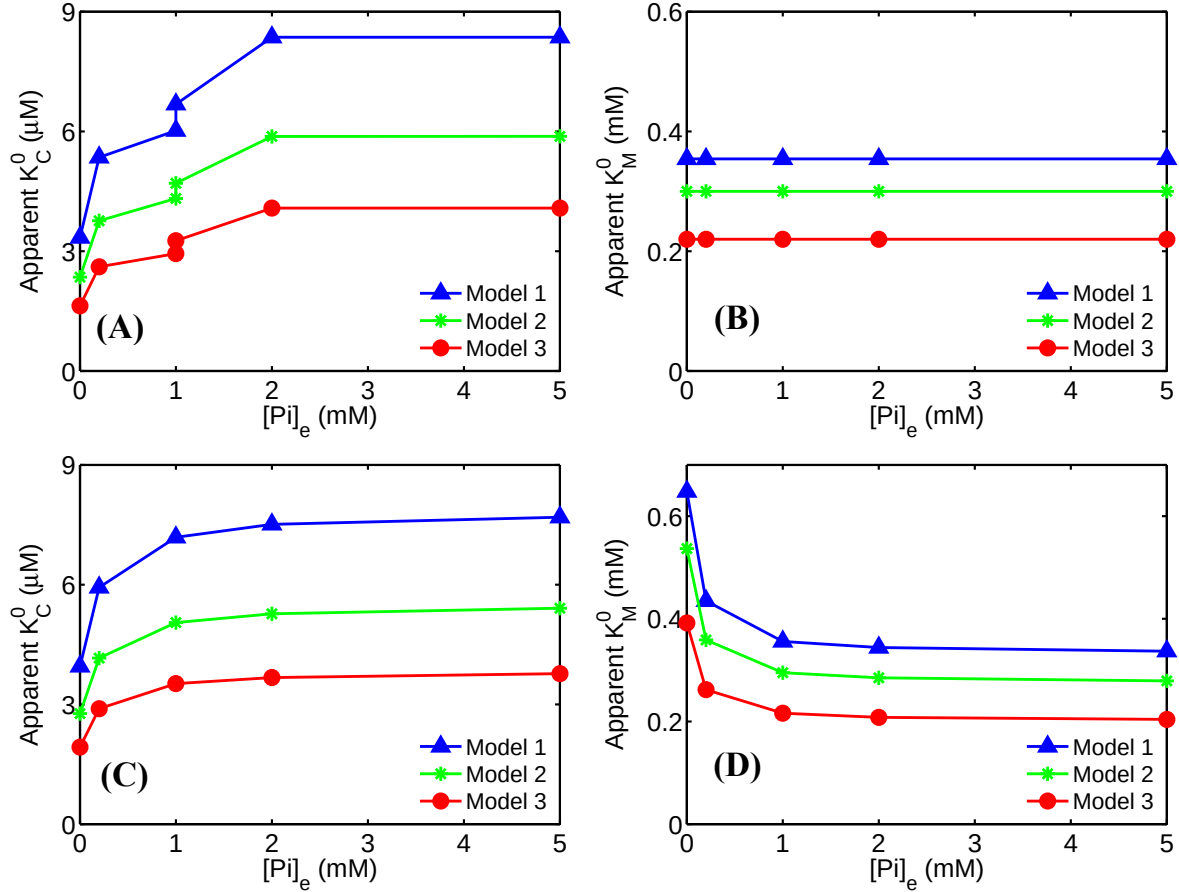
The estimated values of the model parameters under these two different fitting scenarios are summarized in Table S1. Under either scenario, though the model satisfactorily described the available data (similar to those shown in Figures 2 and 3), the estimated  $K_{\text{C}}^0$  (and also  $K_{\text{M}}^0$  in Case B) values were found to differ between different data sets, depending on the amounts of Pi present in the experimental buffer mediums. The variations of the estimated  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  values with the Pi levels under these two scenarios are shown in Figure S2. In the first scenario (Case A), the fitting gives different estimates of  $K_{\text{C}}^0$  values for different data sets obtained from the same source with the same Pi level (e.g., from the work of Crompton et al (5) with  $\text{Pi} = 1 \text{ mM}$ ) (Figure S2A). This questions the viability of the proposed kinetic mechanism of  $\text{Mg}^{2+}$  inhibition of the uniporter function with Pi affecting only the  $K_{\text{C}}^0$  values. On the other hand, in the second scenario (Case B), the estimates of  $K_{\text{C}}^0$  values increase with the Pi levels, while the estimates of  $K_{\text{M}}^0$  values decrease with the Pi levels (i.e.,  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  reciprocally vary with Pi), which are of

the form of Eq. (9). In this case, Pi increases the binding affinity of  $\text{Ca}^{2+}$  and decreases the binding affinity of  $\text{Mg}^{2+}$  for the uniporter. This justifies the proposed Pi dependencies of the kinetic parameters  $K_C^0$  and  $K_M^0$  (Eq. 9) in the present uniporter model.

**Table S1: Estimated parameter values for the kinetic model of mitochondrial  $\text{Ca}^{2+}$  uniporter without considering the Pi regulation mechanism.**

| Parameters                           | Model 1                |                        | Model 2                |                        | Model 3                |                        | Ref. |
|--------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------|
|                                      | Case A                 | Case B                 | Case A                 | Case B                 | Case A                 | Case B                 |      |
| $k^0$                                | $1.26 \times 10^{-3}$  | $1.27 \times 10^{-3}$  | $1.46 \times 10^{-3}$  | $1.48 \times 10^{-3}$  | $1.56 \times 10^{-3}$  | $1.58 \times 10^{-3}$  | (5)  |
|                                      | $20.27 \times 10^{-3}$ | $20.25 \times 10^{-3}$ | $22.02 \times 10^{-3}$ | $22.29 \times 10^{-3}$ | $21.91 \times 10^{-3}$ | $22.20 \times 10^{-3}$ | (3)  |
|                                      | $16.47 \times 10^{-3}$ | $16.45 \times 10^{-3}$ | $19.08 \times 10^{-3}$ | $19.32 \times 10^{-3}$ | $20.34 \times 10^{-3}$ | $20.61 \times 10^{-3}$ | (7)  |
|                                      | $27.87 \times 10^{-3}$ | $27.85 \times 10^{-3}$ | $30.09 \times 10^{-3}$ | $30.46 \times 10^{-3}$ | $28.95 \times 10^{-3}$ | $29.34 \times 10^{-3}$ | (6)  |
|                                      | $63.35 \times 10^{-3}$ | $63.30 \times 10^{-3}$ | $61.65 \times 10^{-3}$ | $62.41 \times 10^{-3}$ | $59.47 \times 10^{-3}$ | $60.27 \times 10^{-3}$ | (4)  |
| $K_C^0$ ([Pi] <sub>e</sub> = 0 mM)   | $3.34 \times 10^{-6}$  | $3.95 \times 10^{-6}$  | $2.35 \times 10^{-6}$  | $2.77 \times 10^{-6}$  | $1.63 \times 10^{-6}$  | $1.93 \times 10^{-6}$  | (5a) |
| $K_C^0$ ([Pi] <sub>e</sub> = 1 mM)   | $6.02 \times 10^{-6}$  | $7.185 \times 10^{-6}$ | $4.23 \times 10^{-6}$  | $5.05 \times 10^{-6}$  | $2.94 \times 10^{-6}$  | $3.52 \times 10^{-6}$  | (5b) |
| $K_C^0$ ([Pi] <sub>e</sub> = 1 mM)   | $6.68 \times 10^{-6}$  | $7.185 \times 10^{-6}$ | $4.7 \times 10^{-6}$   | $5.05 \times 10^{-6}$  | $3.26 \times 10^{-6}$  | $3.52 \times 10^{-6}$  | (5c) |
| $K_C^0$ ([Pi] <sub>e</sub> = 2 mM)   | $8.36 \times 10^{-6}$  | $7.51 \times 10^{-6}$  | $5.875 \times 10^{-6}$ | $5.27 \times 10^{-6}$  | $4.08 \times 10^{-6}$  | $3.67 \times 10^{-6}$  | (3)  |
| $K_C^0$ ([Pi] <sub>e</sub> = 5 mM)   | $8.36 \times 10^{-6}$  | $7.69 \times 10^{-6}$  | $5.875 \times 10^{-6}$ | $5.41 \times 10^{-6}$  | $4.08 \times 10^{-6}$  | $3.77 \times 10^{-6}$  | (7)  |
| $K_C^0$ ([Pi] <sub>e</sub> = 2 mM)   | $8.36 \times 10^{-6}$  | $7.51 \times 10^{-6}$  | $5.875 \times 10^{-6}$ | $5.27 \times 10^{-6}$  | $4.08 \times 10^{-6}$  | $3.67 \times 10^{-6}$  | (6)  |
| $K_C^0$ ([Pi] <sub>e</sub> = 0.2 mM) | $5.35 \times 10^{-6}$  | $5.93 \times 10^{-6}$  | $3.76 \times 10^{-6}$  | $4.16 \times 10^{-6}$  | $2.61 \times 10^{-6}$  | $2.89 \times 10^{-6}$  | (4)  |
| $K_M^0$ ([Pi] <sub>e</sub> = 0 mM)   | $0.354 \times 10^{-3}$ | $0.648 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.537 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.392 \times 10^{-3}$ | (5a) |
| $K_M^0$ ([Pi] <sub>e</sub> = 1 mM)   | $0.354 \times 10^{-3}$ | $0.356 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.295 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.216 \times 10^{-3}$ | (5b) |
| $K_M^0$ ([Pi] <sub>e</sub> = 1 mM)   | $0.354 \times 10^{-3}$ | $0.356 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.295 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.216 \times 10^{-3}$ | (5c) |
| $K_M^0$ ([Pi] <sub>e</sub> = 2 mM)   | $0.354 \times 10^{-3}$ | $0.344 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.285 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.208 \times 10^{-3}$ | (3)  |
| $K_M^0$ ([Pi] <sub>e</sub> = 5 mM)   | $0.354 \times 10^{-3}$ | $0.337 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.279 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.204 \times 10^{-3}$ | (7)  |
| $K_M^0$ ([Pi] <sub>e</sub> = 2 mM)   | $0.354 \times 10^{-3}$ | $0.344 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.285 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.208 \times 10^{-3}$ | (6)  |
| $K_M^0$ ([Pi] <sub>e</sub> = 0.2 mM) | $0.354 \times 10^{-3}$ | $0.435 \times 10^{-3}$ | $0.3 \times 10^{-3}$   | $0.359 \times 10^{-3}$ | $0.22 \times 10^{-3}$  | $0.262 \times 10^{-3}$ | (4)  |
| $\gamma$                             | 3.9                    | 3.9                    | 5.7                    | 5.7                    | 9.2                    | 9.2                    | all  |

The dissociation constants  $K_C^0$  and  $K_M^0$  are in the units of molar concentration, the rate constant  $k^0$  is in the units of nmol/mg/sec, and  $\gamma$  is unitless. The references (5a), (5b) and (5c) correspond to the experimental protocol of Crompton et al (5) with 0 mM Pi and 0 mM  $\text{Mg}^{2+}$ , 1 mM Pi and 0 mM  $\text{Mg}^{2+}$ , and 1 mM Pi and 1 mM  $\text{Mg}^{2+}$  in the buffer medium, respectively.



**Figure S2: Estimated kinetic parameter values as functions of extra-matrix Pi.** Shown are the estimated values of the kinetic parameters  $K_C^0$  and  $K_M^0$  obtained from different experimental data sets with different amounts of Pi present in the experimental buffer mediums. The upper panel (plots A and B) corresponds to the values obtained from the fitting process A (Case A of Table S1), while the lower panel (plots C and D) corresponds to the values obtained from the fitting process B (Case B of Table S1).

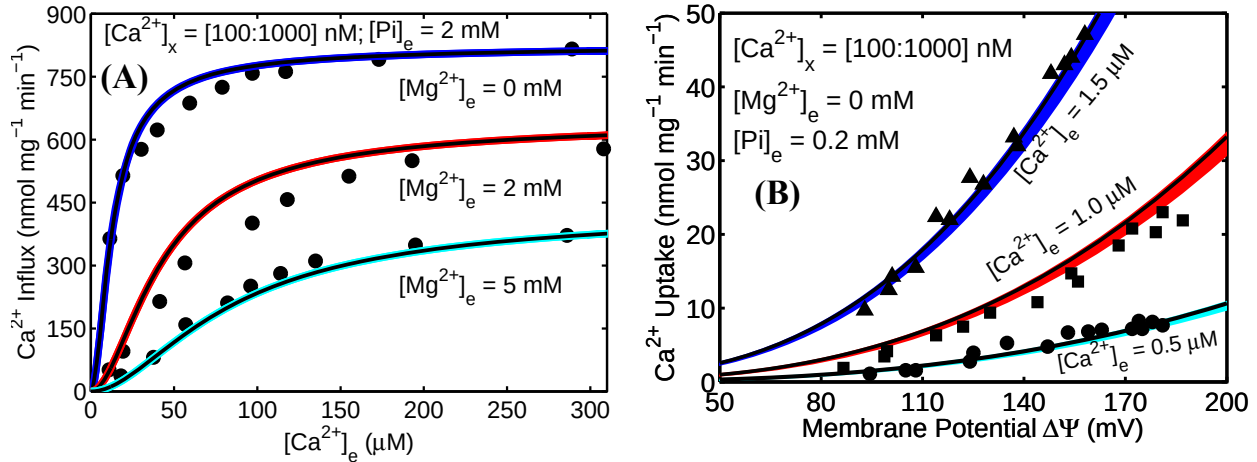
## Appendix SC: Further Model Simulation Analyses of Mitochondrial $Ca^{2+}$ Uptake via the $Ca^{2+}$ Uniporter

This appendix provides further model simulation analyses of the proposed kinetic model of mitochondrial  $Ca^{2+}$  uniporter, describing the nature of the extra-matrix  $Ca^{2+}$  and  $\Delta\Psi$  dependencies and the nature of the  $Mg^{2+}$  inhibition of mitochondrial  $Ca^{2+}$  uptake via the  $Ca^{2+}$  uniporter. For model simulations, Model 2 is selected with the estimated parameter values as described in Table 1. To test the robustness of the model parameters and predications, the mitochondrial  $Ca^{2+}$  uptake were simulated as a function of extra-matrix  $[Ca^{2+}]$  and  $\Delta\Psi$  in response to physiological variations in matrix  $[Ca^{2+}]$  (from 100 nM to 1000 nM) and compared to the experimental data of Bragadin et al (3) and Wingrove et al (4) (Figure S3). It is depicted that, the model is insensitive to variations in matrix  $[Ca^{2+}]$ , predicting a stable physiological operation of the uniporter. The

model predictions corresponding to the other experimental data are also found to be insensitive to variations in matrix  $[Ca^{2+}]$  (not shown).

To predict the saturation of  $Ca^{2+}$  fluxes via the uniporter,  $Ca^{2+}$  uptakes were simulated over a range of extra-matrix  $[Ca^{2+}]$  (0  $\mu M$  to 30  $\mu M$ ) both in the absence and presence of physiological levels of  $Mg^{2+}$  (Figure S4(A,B)), corresponding to the experimental data of Wingrove et al (4). The model simulations suggested that the  $Ca^{2+}$  uptake profiles reach saturation at much lower level of extra-matrix  $[Ca^{2+}]$  ( $\approx 30 \mu M$ ) compared to our previous models of the uniporter ( $\approx 150 \mu M$ ) (Figure S4A). Furthermore, in the presence of 1 mM  $Mg^{2+}$ , the  $Ca^{2+}$  uptake profiles failed to reach saturation at this level of extra-matrix  $[Ca^{2+}]$ , due to the inhibition of mitochondrial  $Ca^{2+}$  uptake by extra-matrix  $Mg^{2+}$  (Figure S4B). A high extra-matrix  $[Ca^{2+}]$  ( $\approx 100 \mu M$ ) is required to reach the saturation (not shown). In addition, to predict the nature of  $Mg^{2+}$  inhibition, mitochondrial  $Ca^{2+}$  uptakes were simulated as a function of extra-matrix  $[Ca^{2+}]$  for a range of  $\Delta\Psi$  (0 mV to 200 mV) both in the absence of  $Mg^{2+}$  (Figure S4C) and presence of 2 mM  $Mg^{2+}$  (Figure S4D), corresponding to the experimental protocol of Bragadin et al (3). The model accurately depicts the nature of  $Mg^{2+}$  inhibition of the uniporter function over a wide range of  $\Delta\Psi$ , observed experimentally (3, 5-7) (i.e., the kinetics is hyperbolic in the absence of  $Mg^{2+}$ , sigmoid in presence of  $Mg^{2+}$ , and the degree of the sigmoidicity increases with increase in  $[Mg^{2+}]$ ).

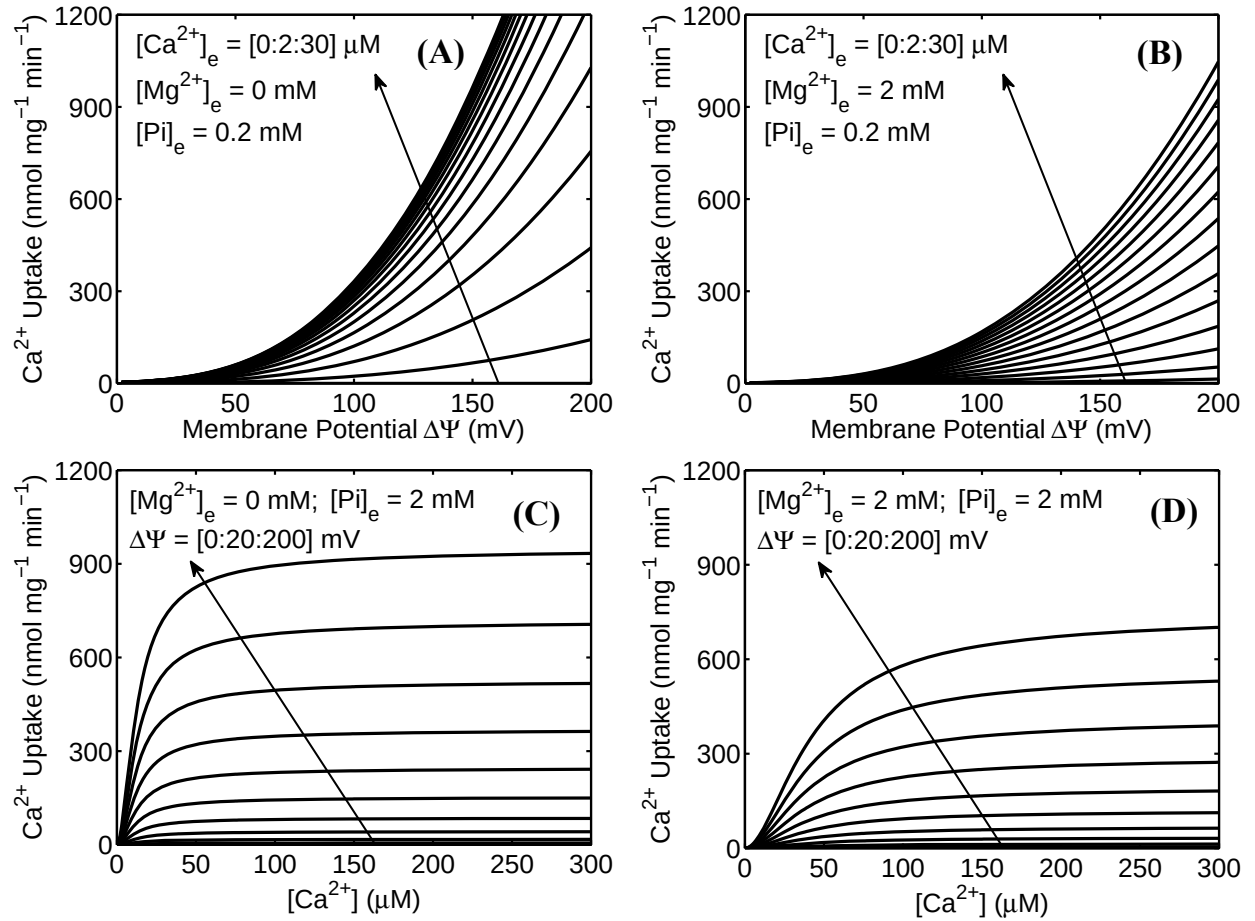
To further predict the nature of  $Mg^{2+}$  inhibition of  $Ca^{2+}$  uptake via the uniporter, the  $Ca^{2+}$  uptakes were also simulated as a function of extra-matrix  $[Mg^{2+}]$  for a range of extra-matrix  $[Ca^{2+}]$  with  $[Pi]_e = 1 \text{ mM}$  (10-190  $\mu M$ ) (Figure S5), corresponding to the experimental protocol of Bragadin et al (3). This simulation is able to provide the amount of  $Mg^{2+}$  required to strongly inhibit mitochondrial  $Ca^{2+}$  uptake via the uniporter for a particular level of extra-matrix  $[Ca^{2+}]$ .



**Figure S3: Predicted sensitivities (color maps) of mitochondrial  $Ca^{2+}$  uptakes as functions of extra-matrix  $[Ca^{2+}]$  and  $\Delta\Psi$  in response to variation in matrix  $[Ca^{2+}]$ , in the presence of different amounts of  $Mg^{2+}$  in the mediums, and their comparisons to the experimental data (points).** For simulations, the matrix  $[Ca^{2+}]$  was varied from 100 nM to 1000 nM for sensitivity color maps. Plot A shows the model simulations and fittings to the kinetic data of Bragadin et al (3) in which the initial rates of  $Ca^{2+}$  influx via the uniporter in response to variations in extra-

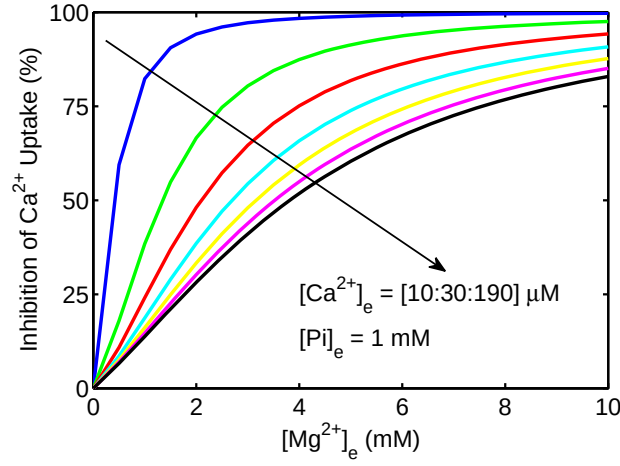


matrix  $[Ca^{2+}]$  were measured in respiring mitochondria of rat livers suspended in the mediums containing different amounts of  $Mg^{2+}$  ( $[Mg^{2+}]_e = 0, 2$  and  $5$  mM) and  $2$  mM Pi. Plot B shows the model simulations and fittings to the kinetic data of Wingrove et al (4) in which the initial rates of  $Ca^{2+}$  influx via the uniporter as a function of  $\Delta\Psi$  were measured in respiring mitochondria of rat livers suspended in mediums with three different levels of extra-matrix  $Ca^{2+}$  ( $[Ca^{2+}]_e = 0.5, 1.0$  and  $1.5$   $\mu M$ ), in the absence of  $Mg^{2+}$  and in the presence of  $0.2$  mM Pi in the mediums. The black lines corresponding to the model fittings to the data are based on  $250$  nM of matrix  $[Ca^{2+}]$ . The matrix  $[Mg^{2+}]$  was set at  $0.4$  mM and the model uses the same parameter values as summarized in Table 1.



**Figure S4: Predictions of mitochondrial  $Ca^{2+}$  uptake as a function of  $\Delta\Psi$  for a range of extra-matrix  $[Ca^{2+}]$  and as a function of extra-matrix  $[Ca^{2+}]$  for a range of  $\Delta\Psi$ , both in the absence and presence of extra-matrix  $Mg^{2+}$ .** The upper Panel (A and B) shows the model predicted  $Ca^{2+}$  uptake as a function of  $\Delta\Psi$  corresponding to the experimental protocol of Wingrove et al (4) with extra-matrix  $[Ca^{2+}]$  ranging from  $0$   $\mu M$  to  $30$   $\mu M$ , both (A) in the absence of  $Mg^{2+}$  and (B) in the presence of  $1$  mM  $Mg^{2+}$ . The lower Panel (C and D) shows the model predicted  $Ca^{2+}$  uptake as a function of extra-matrix  $[Ca^{2+}]$  corresponding to the experimental protocol of Bragadin et al (3), for a range of  $\Delta\Psi$  ( $0$ - $200$  mV) with different amounts of  $Mg^{2+}$  ( $[Mg^{2+}]_e = 0$  and  $2$  mM) present in the buffer mediums. The matrix  $Ca^{2+}$  was fixed at  $250$  nM and matrix  $Mg^{2+}$  was fixed at  $0.4$  mM. The model used the same values of estimated parameters as summa-

rized in Table 1. The arrows indicate the direction of increasing extra-matrix  $[Ca^{2+}]$  in plots A and B and increasing  $\Delta\Psi$  in plots C and D.



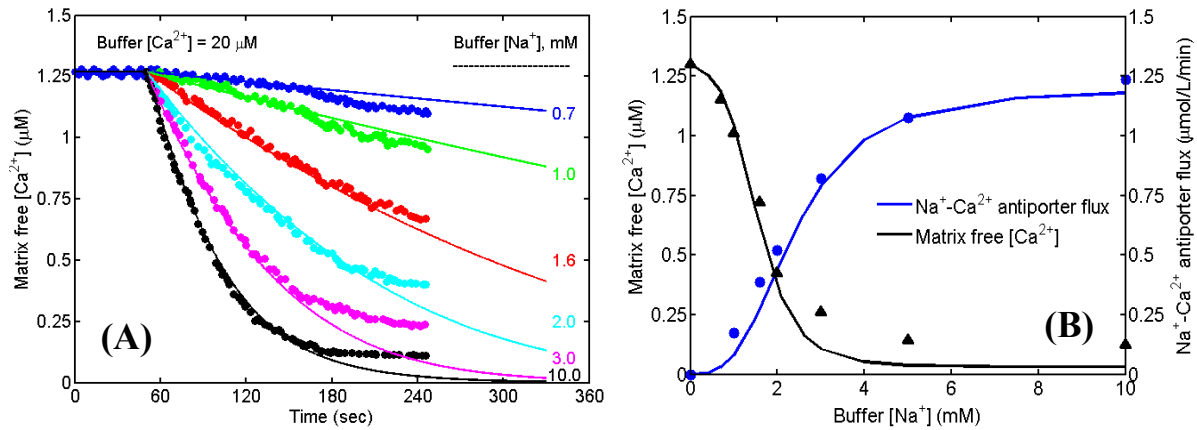
**Figure S5: Predictions of % inhibition of mitochondrial  $Ca^{2+}$  uptake as a function extra-matrix  $[Mg^{2+}]$  for a range of extra-matrix  $[Ca^{2+}]$  with  $[Pi]_e = 1$  mM.** Shown are model predicted % inhibition of mitochondrial  $Ca^{2+}$  uptake as a function of extra-matrix  $[Mg^{2+}]$  for a range of extra matrix  $[Ca^{2+}]$  with  $[Pi]_e = 1$  mM, corresponding to the experimental protocol of Bragadin et al (3). The matrix  $[Ca^{2+}]$  was fixed at 250 nM and matrix  $[Mg^{2+}]$  was fixed at 0.4 mM. The model uses the same values of estimated parameters as summarized in Table 1.

### Appendix SD: Further Validation of the Proposed Model of $Ca^{2+}$ Uniporter by Integrated Modeling of Mitochondrial Bioenergetics and $Ca^{2+}$ Handling

Apart from the initial velocity studies demonstrating the kinetics of mitochondrial  $Ca^{2+}$  uniporter, a large number of kinetic studies demonstrate the changes in mitochondrial free  $[Ca^{2+}]$  in response to physiological variations in extra-matrix  $[Ca^{2+}]$ ,  $[Na^+]$  and  $[Mg^{2+}]$  (8-10). For further validation of the proposed mechanism of  $Mg^{2+}$  inhibition of the uniporter function, we integrated the present model of the uniporter and our recently developed model of a  $nNa^+-1Ca^{2+}$  antiporter (11) to our previous computational model of mitochondrial bioenergetics and  $Ca^{2+}$  handling (12). The integrated model is applied to describe the experimental data of Cox and Matlib (9) in which the dynamics and pseudo-steady state of matrix free  $[Ca^{2+}]$  were measured in isolated respiring mitochondria from rabbit hearts following addition of varying levels of  $Na^+$  to the buffer medium with the uniporter blocked by ruthenium red as well as the experimental data of McCormack et al. (8) in which the pseudo-steady state matrix free  $[Ca^{2+}]$  was measured in isolated respiring mitochondria from rat hearts with addition of different levels of  $Ca^{2+}$  to the buffer medium, both in the presence and absence of different levels of  $Na^+$  and/or  $Mg^{2+}$  in the buffer medium. The integrated model simulations of these experimental data are demonstrated through Figure S6 and D2.

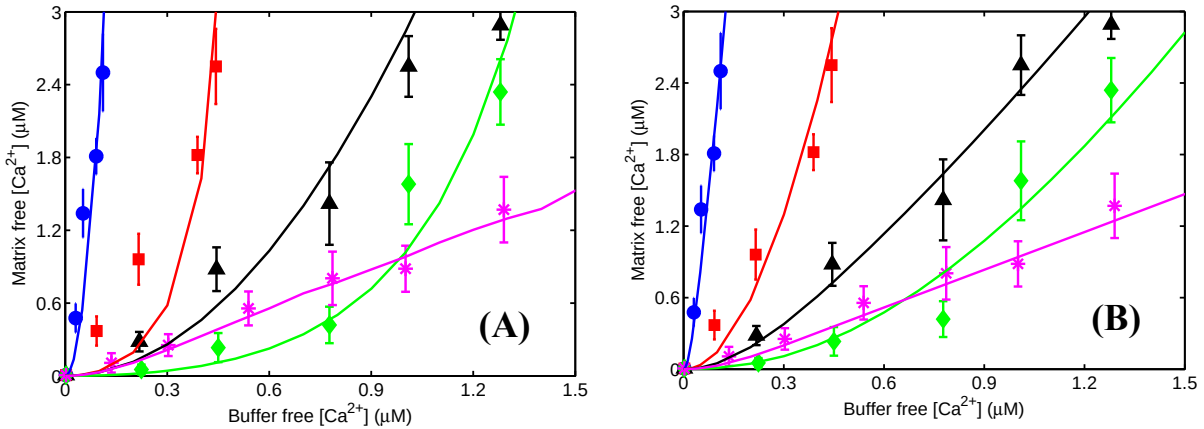
Figure S6 (A) shows the model simulations (lines) of the experimental data (points) of Cox and Matlib (9) on the dynamics of matrix free  $[Ca^{2+}]$  with varying levels of extra-matrix  $Na^+$ . Figure S6 (B) shows the model simulations of the initial rate of decrease of matrix free  $[Ca^{2+}]$

and matrix free  $[Ca^{2+}]$  after 3 min of  $Na^+$  addition for varying levels of extra-matrix  $Na^+$ . Note that, using a  $2Na^+-1Ca^{2+}$  antiporter model ( $n = 2$ ), we are unable to fit these data sets, when simulations are conducted for a range of extra-matrix  $[Ca^{2+}]$  (0-40  $\mu M$ ). However, using a  $3Na^+-1Ca^{2+}$  antiporter model ( $n = 3$ ), we are able to fit data accurately for the entire range of extra-matrix  $[Ca^{2+}]$ . Therefore, a  $3Na^+-1Ca^{2+}$  antiporter model was used for these simulations. This is consistent with our previous analyses of the kinetics and stoichiometry of mitochondrial  $nNa^+-1Ca^{2+}$  antiporter (12). These simulations are reshown here just to revalidate our previous model of the  $nNa^+-1Ca^{2+}$  antiporter (11), which is used later for the simulation of the experimental data of McCormack et al. (8) in which experiments are done with and without different levels of  $Na^+$  (and also  $Mg^{2+}$ ) in the suspending mediums. We note here that the model simulations do not describe the stabilization of the matrix free  $[Ca^{2+}]$  for 2, 3 and 10 mM of  $Na^+$ , in contrast to the data. However, for a  $3Na^+/1Ca^{2+}$  exchange (stoichiometry), all  $Ca^{2+}$  could be extruded from the matrix, when mitochondrial are challenged with high  $[Na^+]$  with fixed matrix  $[Ca^{2+}]$ . Why the model fits the first 2 minutes of these data but not the remaining 30 seconds is unknown. Probably redoing these experiments for longer durations with different amounts of extra-matrix  $Ca^{2+}$  and  $Na^+$  would be helpful to identify fully the kinetics and stoichiometry of the NCE.



**Figure S6: Comparison of the model simulations (lines) to the experimental data (points) of Cox and Matlib (9) on the response of matrix free  $[Ca^{2+}]$  to extra-matrix  $[Na^+]$  perturbations using a  $3Na^+-1Ca^{2+}$  antiporter model.** (A) Shown are the model simulations of the dynamic decay of matrix free  $[Ca^{2+}]$  in response to addition of various levels of  $Na^+$  to the buffer medium. The integrated model uses Model 2 of our previously developed  $3Na^+-1Ca^{2+}$  antiporter model (11) and Model 2 of the present uniporter model. In the antiporter model, the rate constant  $k_i$  and the  $Na^+$  binding constant  $K_{Ne} = K_{Nx} = K_N$  are reestimated to fit the integrated model to this dynamic data, whereas the other model parameters remained the same, as reported before (11). For model simulations, we used  $[Ca^{2+}]_e = 20 \mu M$ , corresponding to a buffer free  $[Ca^{2+}]$  of 1.5  $\mu M$  with 50  $\mu M$  EGTA and buffer pH = 7.2. The uniporter activity was set to zero as per the experimental protocol. The values of  $k_i$  and  $K_N$  are found to be  $2.94 \times 10^{-6}$  nmol/mg/sec and 1.7 mM, respectively, similar order of magnitude as reported before (11). (B) Shown are the simulations of the initial rates of decrease of matrix free  $[Ca^{2+}]$  and matrix free  $[Ca^{2+}]$  in response to variations in extra-matrix  $[Na^+]$  after 3 min of  $Na^+$  addition.

Figure S7 (A) shows the model simulations (lines) of the experimental data (points) of McCormack et al (8) on the pseudo-steady state matrix free  $[Ca^{2+}]$  as a function of extra-matrix  $[Ca^{2+}]$  in the absence or presence of different levels of  $Na^+$  and/or  $Mg^{2+}$  or an uncoupler (FCCP) in the buffer medium. For simulations, the kinetic parameters of the  $3Na^+-1Ca^{2+}$  antiporter model and the present uniporter model were fixed at their same values as estimated before (see Table 1 and Figure S8 legend), whereas the rate constants were allowed to vary to match these data sets. As we were not able to fit the model to these data sets over the entire range of extra-matrix free  $[Ca^{2+}]$ , by considering a specialized inhibition mechanism of the  $Ca^{2+}$  uniporter by matrix free  $[Ca^{2+}]$ , we are able to fit the model accurately to these data sets. The corresponding simulations are shown in Figure S7 (B). This hypothesis is consistent with that is reported in our previous study (12). In this case, we only modify the rate constant of the present uniporter model as:  $k^0 = k^0 \cdot K_i / (K_i + [Ca]_x)$  to account for the inhibition of uniporter activity by matrix free  $[Ca^{2+}]$ . In our previous integrated model of mitochondrial  $Ca^{2+}$  handling (12) both the rate constant  $k^0$  and the  $K_m$  of  $Ca^{2+}$  for the uniporter were modified to fit these data sets and the model provided different kinetic parameters for different data curves. In contrast, the present integrated model is able to simulate accurately the inhibitory effect of  $Mg^{2+}$  on the pseudo-steady state trans-matrix free  $[Ca^{2+}]$  distribution both in the presence and absence of extra-matrix  $[Na^+]$ , with a unique  $K_m$  value of  $Ca^{2+}$  and  $Mg^{2+}$  for the uniporter, validating the proposed mechanism of  $Mg^{2+}$  inhibition.



**Figure S7: Comparison of the model simulations (lines) to the experimental data (points) of McCormack et al (8) on the pseudo-steady state matrix free  $[Ca^{2+}]$  in response to varying levels of extra-matrix  $[Ca^{2+}]$  using the present model of uniporter.** (A) Shown are the model simulations of the pseudo-steady state matrix free  $[Ca^{2+}]$  as a function of extra-matrix  $[Ca^{2+}]$  in the absence or presence of different levels of  $Na^+$  and/or  $Mg^{2+}$  or an uncoupler (FCCP) in the buffer medium. The extra-matrix free  $[Ca^{2+}]$  was maintained at the desired level with the addition of calculated amount of EGTA and  $CaCl_2$  to the buffer medium; blue ● with no further addition, red ■ with 10 mM of  $NaCl$ , black ▲ with 2 mM of  $MgCl_2$ , green ◆ with 10 mM  $NaCl$  and 2 mM  $MgCl_2$  and magenta \* with 1  $\mu M$  of FCCP. For model simulations, the  $Na^+$  and  $Ca^{2+}$  binding constants of the  $3Na^+-1Ca^{2+}$  antiporter model was fixed at same values as estimated before (see Figure S6 legend) and the rate constant  $k^0$  in the antiporter model was found to be  $5.6 \times 10^{-6}$  nmol/mg/sec for all the data sets. The present uniporter model uses the same kinetic parameters as summarized in Table 1 and the rate constant  $k^0$  is found to be: blue ●:  $19.95 \times 10^{-3}$  nmol/mg/sec,

red ■:  $10.64 \times 10^{-3}$  nmol/mg/sec, black ▲:  $0.266 \times 10^{-3}$  nmol/mg/sec, green ♦:  $1.064 \times 10^{-3}$  nmol/mg/sec and magenta ●:  $0.023$  nmol/mg/sec. (B) Shown are the corresponding model simulations with the specialized inhibition mechanism of the  $\text{Ca}^{2+}$  uniporter by matrix free  $[\text{Ca}^{2+}]$ . In this case, the modified rate constant  $k^{0'}$  of the uniporter is found to be: blue ●:  $40 \times 10^{-3}$  nmol/mg/sec, red ■:  $40 \times 10^{-3}$  nmol/mg/sec, black ▲:  $0.46 \times 10^{-3}$  nmol/mg/sec, green ♦:  $2.92 \times 10^{-3}$  nmol/mg/sec and magenta \*:  $186.2$  nmol/mg/sec.

## Appendix SE: Alternate (Reduced) Kinetic Models for $\text{Mg}^{2+}$ Inhibition of Mitochondrial $\text{Ca}^{2+}$ Uptake via the $\text{Ca}^{2+}$ Uniporter

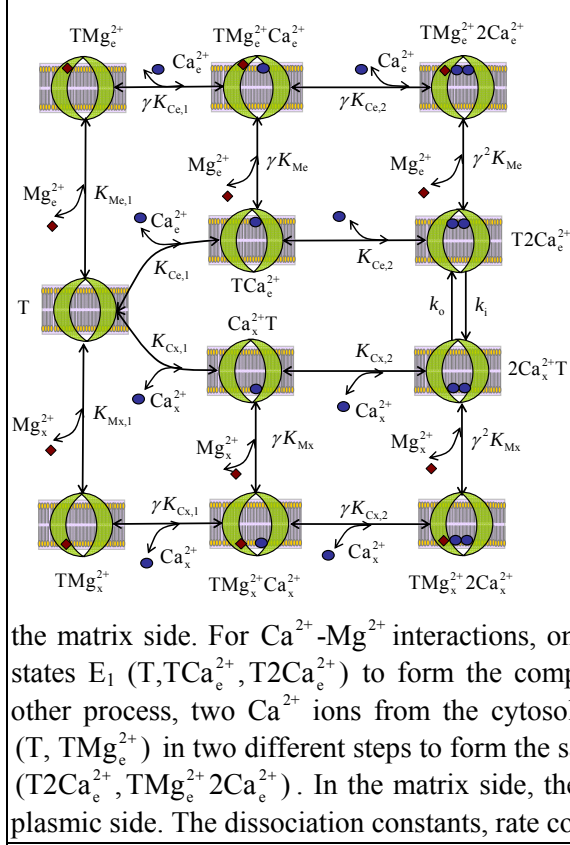
This appendix presents two alternate (reduced) kinetic models of mitochondrial  $\text{Ca}^{2+}$  uniporter based on two simpler kinetic schemes of  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake via the uniporter (see Figure S8 and E2). The uniporter flux expressions ( $J_{\text{CU}}$ ) can be derived using similar approach as described in Appendix SA (see below). We followed a similar systematic fitting procedure using multiple steps to parameterize these reduced uniporter models. The estimated values of the model parameters with the Pi regulation mechanisms for the kinetic parameters  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  incorporated are summarized in Table S2, and the comparisons of the model outputs to the experimental data are illustrated in Figures E3 and E4. From the model analyses, it is apparent that none of these reduced models is able to describe the available experimental data satisfactorily, as that of the proposed model of the uniporter (Figures 2 and 3). To best fit these reduced models to the available data, the parameters  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  have to be different for different data sets, even with the proposed Pi regulation mechanisms for the parameters  $K_{\text{C}}^0$  and  $K_{\text{M}}^0$  incorporated. Without these Pi regulation mechanisms, the model further deviates from the available data that is shown in Figures E3 and E4. Therefore, irrespective of the Pi dependencies of the kinetic parameters, these simpler models cannot describe the observed kinetics of  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake via the  $\text{Ca}^{2+}$  uniporter.

**Reduced Kinetic Scheme-1:** Based on the reduced kinetic scheme-1 for  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake shown in Figure S6, the net  $\text{Ca}^{2+}$  transport flux via the uniporter ( $J_{\text{CU}}$ ) is given by Eq. (A7) in which the denominator  $D$  is given by

$$D = D_1 = 1 + \frac{[\text{Ca}^{2+}]_{\text{e}}^2}{K_{\text{Ce}}^2} + \frac{[\text{Ca}^{2+}]_{\text{x}}^2}{K_{\text{Cx}}^2} + \frac{[\text{Mg}^{2+}]_{\text{e}}}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_{\text{x}}}{K_{\text{Mx}}} + \frac{[\text{Ca}^{2+}]_{\text{e}}^2 [\text{Mg}^{2+}]_{\text{e}}}{\gamma^2 \cdot K_{\text{Ce}}^2 \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_{\text{x}}^2 [\text{Mg}^{2+}]_{\text{x}}}{\gamma^2 \cdot K_{\text{Cx}}^2 \cdot K_{\text{Mx}}} \quad (\text{Model 1}), \quad (\text{E1a})$$

$$D = D_2 = 1 + \frac{[\text{Ca}^{2+}]_{\text{e}}}{K_{\text{Ce}}} + \frac{[\text{Ca}^{2+}]_{\text{e}}^2}{K_{\text{Ce}}^2} + \frac{[\text{Ca}^{2+}]_{\text{x}}}{K_{\text{Cx}}} + \frac{[\text{Ca}^{2+}]_{\text{x}}^2}{K_{\text{Cx}}^2} + \frac{[\text{Mg}^{2+}]_{\text{e}}}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_{\text{x}}}{K_{\text{Mx}}} + \frac{[\text{Ca}^{2+}]_{\text{e}} [\text{Mg}^{2+}]_{\text{e}}}{\gamma \cdot K_{\text{Ce}} \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_{\text{e}}^2 [\text{Mg}^{2+}]_{\text{e}}}{\gamma^2 \cdot K_{\text{Ce}}^2 \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_{\text{x}} [\text{Mg}^{2+}]_{\text{x}}}{\gamma \cdot K_{\text{Cx}} \cdot K_{\text{Mx}}} + \frac{[\text{Ca}^{2+}]_{\text{x}}^2 [\text{Mg}^{2+}]_{\text{x}}}{\gamma^2 \cdot K_{\text{Cx}}^2 \cdot K_{\text{Mx}}} \quad (\text{Model 2}), \quad (\text{E1b})$$

$$D = D_3 = 1 + 2 \frac{[\text{Ca}^{2+}]_{\text{e}}}{K_{\text{Ce}}} + \frac{[\text{Ca}^{2+}]_{\text{e}}^2}{K_{\text{Ce}}^2} + 2 \frac{[\text{Ca}^{2+}]_{\text{x}}}{K_{\text{Cx}}} + \frac{[\text{Ca}^{2+}]_{\text{x}}^2}{K_{\text{Cx}}^2} + \frac{[\text{Mg}^{2+}]_{\text{e}}}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_{\text{x}}}{K_{\text{Mx}}} + 2 \frac{[\text{Ca}^{2+}]_{\text{e}} [\text{Mg}^{2+}]_{\text{e}}}{\gamma \cdot K_{\text{Ce}} \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_{\text{e}}^2 [\text{Mg}^{2+}]_{\text{e}}}{\gamma^2 \cdot K_{\text{Ce}}^2 \cdot K_{\text{Me}}} + 2 \frac{[\text{Ca}^{2+}]_{\text{x}} [\text{Mg}^{2+}]_{\text{x}}}{\gamma \cdot K_{\text{Cx}} \cdot K_{\text{Mx}}} + \frac{[\text{Ca}^{2+}]_{\text{x}}^2 [\text{Mg}^{2+}]_{\text{x}}}{\gamma^2 \cdot K_{\text{Cx}}^2 \cdot K_{\text{Mx}}} \quad (\text{Model 3}). \quad (\text{E1c})$$



**Figure S8: Alternate kinetic scheme-1 for  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake via the  $\text{Ca}^{2+}$  uniporter.** This scheme is a reduced version of the proposed kinetic scheme for  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake shown in Figure S1. In this scheme, the uniporter functional unit is assumed to have two binding sites for  $\text{Ca}^{2+}$ , but only one distinct allosteric site for  $\text{Mg}^{2+}$ , facing either the cytosolic side or the matrix side depending on the state of the uniporter. In one process, two  $\text{Ca}^{2+}$  ions from the cytosolic side cooperatively bind to the uniporter T in two different steps to form the complexes  $\text{TCa}_e^{2+}$  and  $\text{T2Ca}_e^{2+}$ . The complex  $\text{T2Ca}_e^{2+}$  then undergoes conformational changes to form the complex  $2\text{Ca}_x^{2+}\text{T}$  in the matrix side. In a reverse process, the complex  $2\text{Ca}_x^{2+}\text{T}$  dissociates in two different steps to form the complex  $\text{Ca}_x^{2+}\text{T}$ , unbound uniporter T, and two free  $\text{Ca}^{2+}$  ions in

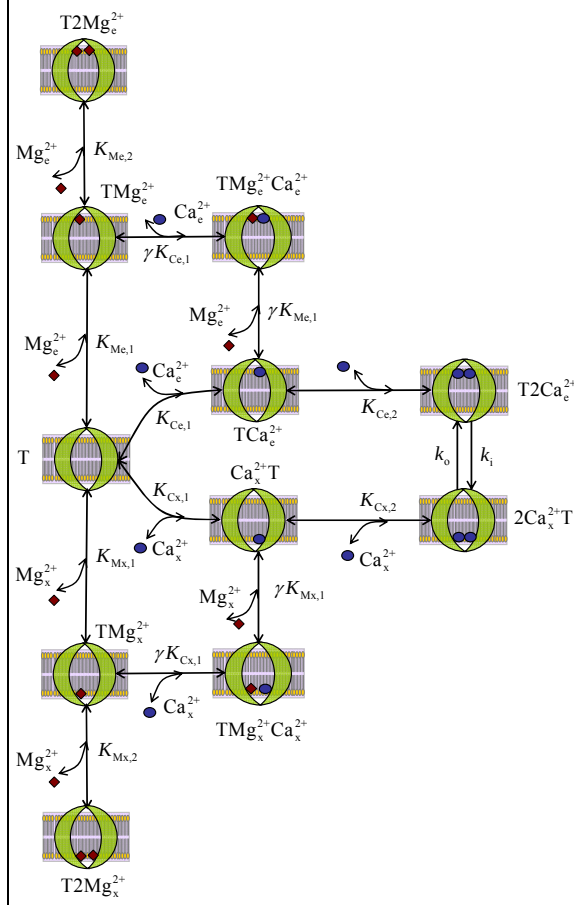
the matrix side. For  $\text{Ca}^{2+}$ - $\text{Mg}^{2+}$  interactions, one  $\text{Mg}^{2+}$  ion from the cytosolic side binds to the uniporter states  $E_1$  ( $\text{T}, \text{TCa}_e^{2+}, \text{T2Ca}_e^{2+}$ ) to form the complexes  $E_1\text{Mg}_e^{2+}$  ( $\text{TMg}_e^{2+}, \text{TMg}_e^{2+}\text{Ca}_e^{2+}, \text{TMg}_e^{2+}2\text{Ca}_e^{2+}$ ). In another process, two  $\text{Ca}^{2+}$  ions from the cytosolic side can cooperatively bind to the uniporter states  $E_2$  ( $\text{T}, \text{TMg}_e^{2+}$ ) in two different steps to form the same complexes  $E_2\text{Ca}_e^{2+}$  ( $\text{TCa}_e^{2+}, \text{TMg}_e^{2+}\text{Ca}_e^{2+}$ ) and  $E_22\text{Ca}_e^{2+}$  ( $\text{T2Ca}_e^{2+}, \text{TMg}_e^{2+}2\text{Ca}_e^{2+}$ ). In the matrix side, the  $\text{Ca}^{2+}$ - $\text{Mg}^{2+}$  interaction is symmetrical to that in the cytoplasmic side. The dissociation constants, rate constants, and  $\gamma$  are as defined in Figure S1.

**Reduced Kinetic Scheme-2:** Based on the reduced kinetic scheme-2 for  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake shown in Figure S9, the net  $\text{Ca}^{2+}$  transport flux via the uniporter ( $J_{\text{CU}}$ ) is given by Eq. (A7) in which the denominator  $D$  is given by

$$D = D_1 = 1 + \frac{[\text{Ca}^{2+}]_e^2}{K_{\text{Ce}}^2} + \frac{[\text{Ca}^{2+}]_x^2}{K_{\text{Cx}}^2} + \frac{[\text{Mg}^{2+}]_e^2}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_x^2}{K_{\text{Mx}}} \quad (\text{Model 1}), \quad (\text{E2a})$$

$$D = D_2 = 1 + \frac{[\text{Ca}^{2+}]_e}{K_{\text{Ce}}} + \frac{[\text{Ca}^{2+}]_e^2}{K_{\text{Ce}}^2} + \frac{[\text{Ca}^{2+}]_x}{K_{\text{Cx}}} + \frac{[\text{Ca}^{2+}]_x^2}{K_{\text{Cx}}^2} + \frac{[\text{Mg}^{2+}]_e}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_e^2}{K_{\text{Me}}^2} + \frac{[\text{Mg}^{2+}]_x}{K_{\text{Mx}}} + \frac{[\text{Mg}^{2+}]_x^2}{K_{\text{Mx}}^2} + \frac{[\text{Ca}^{2+}]_e[\text{Mg}^{2+}]_e}{\gamma \cdot K_{\text{Ce}} \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_x[\text{Mg}^{2+}]_x}{\gamma \cdot K_{\text{Cx}} \cdot K_{\text{Mx}}} \quad (\text{Model 2}), \quad (\text{E2b})$$

$$D = D_3 = 1 + 2 \frac{[\text{Ca}^{2+}]_e}{K_{\text{Ce}}} + \frac{[\text{Ca}^{2+}]_e^2}{K_{\text{Ce}}^2} + 2 \frac{[\text{Ca}^{2+}]_x}{K_{\text{Cx}}} + \frac{[\text{Ca}^{2+}]_x^2}{K_{\text{Cx}}^2} + 2 \frac{[\text{Mg}^{2+}]_e}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_e^2}{K_{\text{Me}}^2} + 2 \frac{[\text{Mg}^{2+}]_x}{K_{\text{Mx}}} + \frac{[\text{Mg}^{2+}]_x^2}{K_{\text{Mx}}^2} + 4 \frac{[\text{Ca}^{2+}]_e[\text{Mg}^{2+}]_e}{\gamma \cdot K_{\text{Ce}} \cdot K_{\text{Me}}} + 4 \frac{[\text{Ca}^{2+}]_x[\text{Mg}^{2+}]_x}{\gamma \cdot K_{\text{Cx}} \cdot K_{\text{Mx}}} \quad (\text{Model 3}), \quad (\text{E2c})$$



**Figure S9: Alternate kinetic scheme-2 for  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  transport via the  $\text{Ca}^{2+}$  uniporter.** This scheme is another reduced version of the proposed kinetic scheme for  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake shown in Figure S1. In this scheme, the uniporter functional unit is assumed to have only two binding sites for  $\text{Ca}^{2+}$ , facing either the cytosolic side or the matrix side depending on the state of the uniporter.  $\text{Mg}^{2+}$  is assumed to compete for the  $\text{Ca}^{2+}$  binding sites. In one process, two  $\text{Ca}^{2+}$  ions from the cytosolic side cooperatively bind to the uniporter T in two different steps to form the complexes  $\text{TCa}_e^{2+}$  and  $\text{T2Ca}_e^{2+}$ . The complex  $\text{T2Ca}_e^{2+}$  then undergoes conformational changes to form the complex  $2\text{Ca}_x^{2+}\text{T}$  in the matrix side. In a reverse process, the complex  $2\text{Ca}_x^{2+}\text{T}$  dissociates in two different steps to form the complex  $\text{Ca}_x^{2+}\text{T}$ , unbound uniporter T, and two free  $\text{Ca}^{2+}$  ions in the matrix side. For  $\text{Ca}^{2+}$ - $\text{Mg}^{2+}$  interactions, two  $\text{Mg}^{2+}$  ions from the cytosolic side cooperatively bind to the uniporter T in two distinct steps to form the complexes  $\text{TMg}_e^{2+}$  and  $\text{T2Mg}_e^{2+}$ . In another process, one  $\text{Ca}^{2+}$  ion can bind to the complex  $\text{TMg}_e^{2+}$  or one  $\text{Mg}^{2+}$  ion can bind to the complex  $\text{TCa}_e^{2+}$  to form the complex  $\text{TMg}_e^{2+}\text{Ca}_e^{2+}$ . In the matrix side, the  $\text{Ca}^{2+}$ - $\text{Mg}^{2+}$  interaction is symmetrical to that in the cytoplasmic side. This

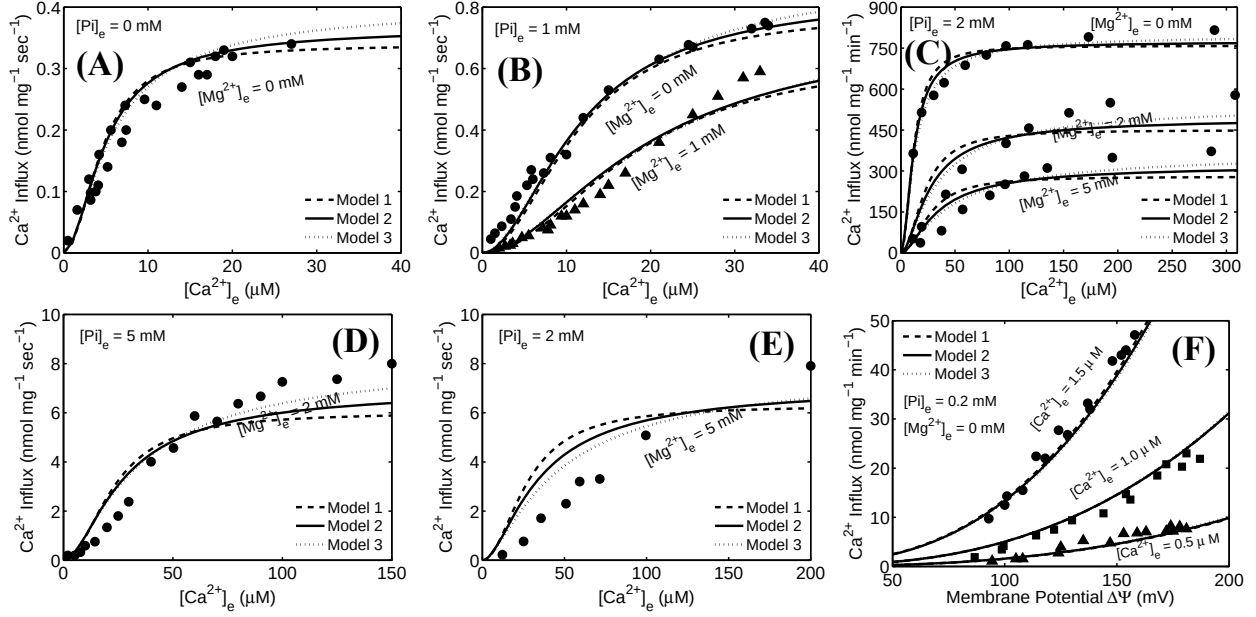
kinetic scheme is purely a competitive inhibition scheme  $\text{Mg}^{2+}$  inhibition of mitochondrial  $\text{Ca}^{2+}$  uptake. The dissociation constants, rate constants, and  $\gamma$  are as defined in Figure S1.

**Table S2: Estimated parameter values for the kinetic models of mitochondrial  $\text{Ca}^{2+}$  uniporter based on two alternate (reduced) kinetic schemes of  $\text{Mg}^{2+}$  inhibition.**

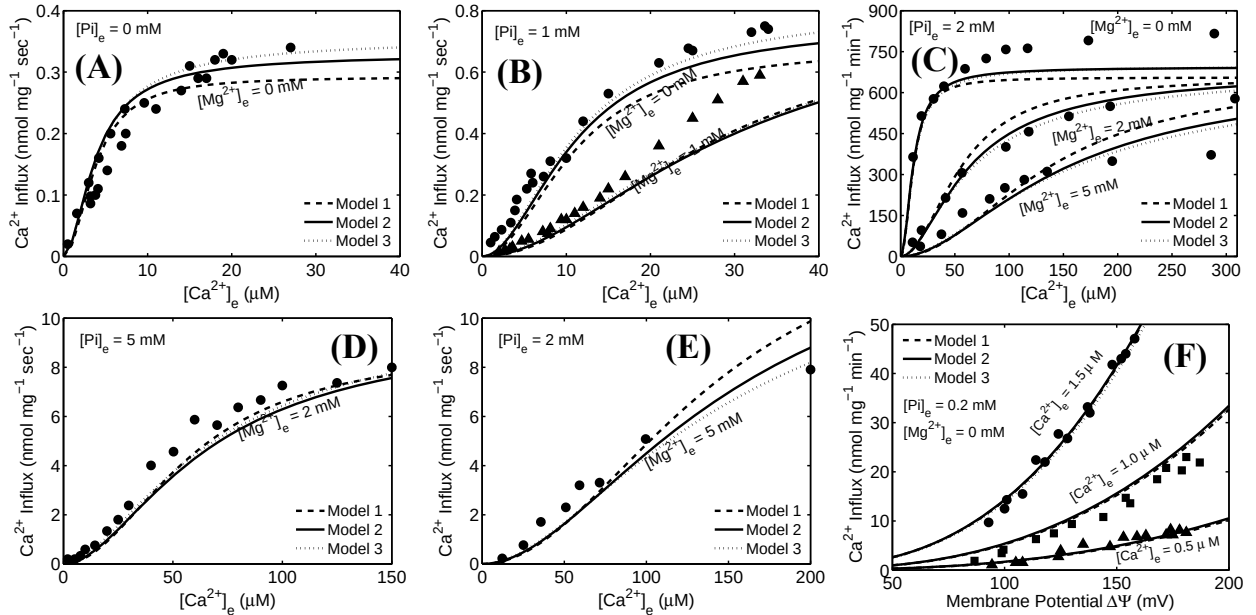
| Parameters          | Reduced Kinetic Scheme 1 |                        |                        | Reduced Kinetic Scheme 2 |                        |                        | Ref. |
|---------------------|--------------------------|------------------------|------------------------|--------------------------|------------------------|------------------------|------|
|                     | Model 1                  | Model 2                | Model 3                | Model 1                  | Model 2                | Model 3                |      |
| $k^0$ (nmol/mg/sec) | $1.32 \times 10^{-3}$    | $1.44 \times 10^{-3}$  | $1.59 \times 10^{-3}$  | $1.14 \times 10^{-3}$    | $1.29 \times 10^{-3}$  | $1.39 \times 10^{-3}$  | (5)  |
|                     | $21.07 \times 10^{-3}$   | $21.73 \times 10^{-3}$ | $22.28 \times 10^{-3}$ | $18.24 \times 10^{-3}$   | $19.35 \times 10^{-3}$ | $19.46 \times 10^{-3}$ | (3)  |
|                     | $17.12 \times 10^{-3}$   | $18.83 \times 10^{-3}$ | $20.69 \times 10^{-3}$ | $14.82 \times 10^{-3}$   | $16.77 \times 10^{-3}$ | $18.07 \times 10^{-3}$ | (7)  |
|                     | $28.97 \times 10^{-3}$   | $29.70 \times 10^{-3}$ | $29.45 \times 10^{-3}$ | $25.08 \times 10^{-3}$   | $26.44 \times 10^{-3}$ | $25.71 \times 10^{-3}$ | (6)  |
|                     | $65.85 \times 10^{-3}$   | $60.86 \times 10^{-3}$ | $60.48 \times 10^{-3}$ | $57.01 \times 10^{-3}$   | $54.18 \times 10^{-3}$ | $52.82 \times 10^{-3}$ | (4)  |
| $K_C^0$ (M)         | $0.81 \times 10^{-6}$    | $1.44 \times 10^{-6}$  | $1.60 \times 10^{-6}$  | $2.31 \times 10^{-6}$    | $0.97 \times 10^{-6}$  | $0.83 \times 10^{-6}$  | all  |
| $K_M^0$ (M)         | $0.013 \times 10^{-3}$   | $0.050 \times 10^{-3}$ | $0.068 \times 10^{-3}$ | $0.309 \times 10^{-3}$   | $0.139 \times 10^{-3}$ | $0.130 \times 10^{-3}$ | all  |
| $\gamma$ (Unitless) | 20.7                     | 11.7                   | 11.0                   | N/A                      | 2.04                   | 5.2                    | all  |
| $K_{Pi}$ (M)        | $0.2 \times 10^{-3}$     | $0.2 \times 10^{-3}$   | $0.2 \times 10^{-3}$   | $0.2 \times 10^{-3}$     | $0.2 \times 10^{-3}$   | $0.2 \times 10^{-3}$   | all  |
| $E_p$               | 0.199                    | 0.140                  | 0.101                  | 0.19                     | 0.140                  | 0.116                  | all  |

$E_p$  is the combined least square error at the optimal parameter estimates; N/A is not applicable.





**Figure S10: Comparison of the alternate (reduced kinetic scheme-1)  $\text{Ca}^{2+}$  uniporter models (lines) to the experimental data (points) on the extra-matrix  $\text{Ca}^{2+}$  and  $\Delta\Psi$  dependencies of mitochondrial  $\text{Ca}^{2+}$  uptake measured in the absence and presence of extra-matrix  $\text{Mg}^{2+}$ .** Shown are the fittings of three different versions of the model (Model 1, Model 2, and Model 3) to the kinetic data of Crompton et al (5) (plots A, B), Bragadin et al (3) (plot C), Vinogradov and Scarpa (7) (plot D), Scarpa and Graziotti (6) (plot E), and Wingrove et al (4) (plot F).



**Figure S11: Comparison of the alternate (reduced kinetic scheme-2)  $\text{Ca}^{2+}$  uniporter models (lines) to the experimental data (points) on the extra-matrix  $\text{Ca}^{2+}$  and  $\Delta\Psi$  dependencies of mitochondrial  $\text{Ca}^{2+}$  uptake measured in the absence and presence of extra-matrix  $\text{Mg}^{2+}$ .** Shown are the fittings of three different versions of the model (Model 1, Model 2, and Model 3)



to the kinetic data of Crompton et al (5) (plots A, B), Bragadin et al (3) (plot C), Vinogradov and Scarpa (7) (plot D), Scarpa and Graziotti (6) (plot E), and Wingrove et al (4) (plot F).

## References:

1. Dash, R.K., F. Qi, and D.A. Beard. 2009. A biophysically based mathematical model for the kinetics of mitochondrial calcium uniporter. *Biophys J* 96(4):1318-1332.
2. Pradhan, R.K., F. Qi, D.A. Beard, and R.K. Dash. Characterization of membrane potential dependency of mitochondrial  $\text{Ca}^{2+}$  uptake by an improved biophysical model of mitochondrial  $\text{Ca}^{2+}$  uniporter. *PLoS One* 5(10):e13278.
3. Bragadin, M., T. Pozzan, and G.F. Azzone. 1979. Kinetics of  $\text{Ca}^{2+}$  carrier in rat liver mitochondria. *Biochemistry* 18(26):5972-5978.
4. Wingrove, D.E., J.M. Amatruda, and T.E. Gunter. 1984. Glucagon effects on the membrane potential and calcium uptake rate of rat liver mitochondria. *J Biol Chem* 259(15):9390-9394.
5. Crompton, M., E. Sigel, M. Salzmänn, and E. Carafoli. 1976. A kinetic study of the energy-linked influx of  $\text{Ca}^{2+}$  into heart mitochondria. *Eur J Biochem* 69:429-434.
6. Scarpa, A., and P. Graziotti. 1973. Mechanisms for intracellular calcium regulation in heart. I. Stopped-flow measurements of  $\text{Ca}^{2+}$  uptake by cardiac mitochondria. *J Gen Physiol* 62(6):756-772.
7. Vinogradov, A., and A. Scarpa. 1973. The initial velocities of calcium uptake by rat liver mitochondria. *J Biol Chem* 248(15):5527-5531.
8. McCormack, J.G., H.M. Browne, and N.J. Dawes. 1989. Studies on mitochondrial  $\text{Ca}^{2+}$ -transport and matrix  $\text{Ca}^{2+}$  using fura-2-loaded rat heart mitochondria. *Biochim Biophys Acta* 973(3):420-427.
9. Cox, D.A., and M.A. Matlib. 1993. A role for the mitochondrial  $\text{Na}^{+}$ - $\text{Ca}^{2+}$  exchanger in the regulation of oxidative phosphorylation in isolated heart mitochondria. *J Biol Chem* 268(2):938-947.
10. Wan, B., K.F. LaNoue, J.Y. Cheung, and R.C. Scaduto, Jr. 1989. Regulation of citric acid cycle by calcium. *J Biol Chem* 264(23):13430-13439.
11. Pradhan, R.K., D.A. Beard, and R.K. Dash. 2010. A biophysically-based mathematical model for the kinetics of mitochondrial  $\text{Na}^{+}$ - $\text{Ca}^{2+}$  antiporter. *Biophys J* 98(2):1-13.
12. Dash, R.K., and D.A. Beard. 2008. Analysis of cardiac mitochondrial  $\text{Na}^{+}$ - $\text{Ca}^{2+}$  exchanger kinetics with a biophysical model of mitochondrial  $\text{Ca}^{2+}$  handling suggests a 3:1 stoichiometry. *J Physiol* 586(13):3267-3285.